

Programming Fundamentals Mid-I Exam

(CS1002)

Date: 21st September 2026

Course Instructor(s)

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Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Roll No

Section

Student Signature

Attempt all the questions.

CLO 2: Examine code writing, compiling, debugging and program execution.

Question No. 1: Identify any logical or syntactical errors in the following code. Correct the errors, if any, and write the output produced by the corrected code. [15 Min, 2*5=10 Marks]

<pre>1.int main() { int i = 2, j = 3, k, l ; float a, b ; k = i / j * j ; l = j / i * i ; a = i / j * j ; b = j / i * i ; a++; b++; printf("%d %d %d %d", k, l, a, b) ; }</pre> Output 0 2 1.000000 3.000000	<pre>2.int main() { int marks = 75; switch (marks / 10) { case 7.5: printf("Grade B\n"); break; case 7: printf("Grade B+\n"); break; default: printf("Grade Unknown\n"); } }</pre> Output: Grade B+
<pre>3. int main() { int a = 10, b = 20, c = 15; if (a > b) { if (a > c); printf("a is greatest\n") } else printf("c is greatest\n"); }</pre> Output: b is greatest	<pre>4. int main() { int a = 5, b = 0, c = 3; if (a > b && b == c a < 10) { printf("Condition true\n"); } else { printf("Condition false\n"); } return 0; }</pre> Output: Condition true
<pre>5. int main() { int num = 10; if (num % 2 == 0); { printf("Even number\n"); } }</pre>	

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```
else
    printf("Odd number\n");
}
```

Output:

Even number

CLO 1: Describe fundamental concepts of structured and procedural programming, use pseudo- codes and simple programs to understand control structures, iterative structures and functions using C language.

Question No. 2

Write a C program that takes the required input, performs the calculations, checks the given conditions, and displays the appropriate message for each condition. **[20 Min, Total Marks 10]**

A courier company uses a Smart Dispatch System to decide how a package should be handled before delivery. The system receives the following information as input from the user: **[3.5 Marks]**

- Package ID should take numeric value of any size digit as an input for example **1234**
- Package weight in kg should take numeric value which can also be decimal for example **5.5 KG**
- Customer type should take the choice from customer which includes:
 - 1 = Regular
 - 2 = Premium
- Delivery zone:
 - 1 = Local
 - 2 = National
 - 3 = International
- Take two spinner readings as an input from the user. It should take the numeric value between **2 and 9**
- Control Value should take the numeric value. This value will be used together with the last digit of the Package ID to calculate the dispatch limit.

The system performs the following calculations: **[3.5 Marks]**

1. The last digit of the Package ID is extracted.
 2. The system adds this digit to a control value and calculates the dispatch limit.
 3. The two spinner readings are then used to calculate a dispatch score:
 - If both readings are equal, the score is 10 times the reading.
 - Otherwise, the score is the product of the two readings.
 4. If the sum of the two spinner readings is equal to the dispatch limit, the package is marked for Manual Inspection.
 5. If the package is not marked for inspection, determine its delivery priority using these rules: **[3 Marks]**
- International packages
 - If the Weight is above 20 kg, display “**Restricted**”
 - If the score is 50 or above, display “**High priority**”.
 - Else, display “**Standard**”
 - National packages
 - If the customer is Premium and the score is 40 or above, display “**High Priority**”.
 - If the customer is Regular and score is 50 or above, display “**High Priority**”.
 - Else, display “**Standard**”
 - Local packages
 - If the Weight is 5 kg or less and score is 30 or above, display “**Express**”.
 - Else display “**Standard**”
 - If the package is Premium and has a score of 60 or above, an additional message should be displayed: “**Premium Priority Upgrade**”.

Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int packageID;
```

```
    float weight;
```

```
    int customerType, zone;
```

```
    int spinner1, spinner2;
```

```
    int controlValue;
```

```
    int lastDigit, dispatchLimit;
```

```
    int score, sum;
```

```
    printf("Enter Package ID: ");
```

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```
scanf("%d", &packageID);
printf("Enter Package Weight (kg): ");
scanf("%f", &weight);
printf("Enter Customer Type (1-Regular, 2-Premium): ");
scanf("%d", &customerType);
printf("Enter Delivery Zone (1-Local, 2-National, 3-International): ");
scanf("%d", &zone);
printf("Enter first spinner reading (2-9): ");
scanf("%d", &spinner1);
printf("Enter second spinner reading (2-9): ");
scanf("%d", &spinner2);
printf("Enter Control Value: ");
scanf("%d", &controlValue);
lastDigit = packageID % 10;
dispatchLimit = lastDigit + controlValue;
if (spinner1 == spinner2)
{
    score = 10 * spinner1;
}
else
{
    score = spinner1 * spinner2;
}
sum = spinner1 + spinner2;

printf("\nDispatch Limit = %d\n", dispatchLimit);
printf("Dispatch Score = %d\n", score);
if (sum == dispatchLimit)
{
    printf("Manual Inspection\n");
}
else
{
    if (zone == 3)
    {
        if (weight > 20)
        {
            printf("Restricted\n");
        }
        else if (score >= 50)
        {
            printf("High Priority\n");
        }
        else
        {
            printf("Standard\n");
        }
    }
    else if (zone == 2)
    {
        if (customerType == 2 && score >= 40)
        {
            printf("High Priority\n");
        }
        else if (customerType == 1 && score >= 50)
        {
            printf("High Priority\n");
        }
        else
        {
            printf("Standard\n");
        }
    }
}
```

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```
    }  
  }  
  else if (zone == 1)  
  {  
    if (weight <= 5 && score >= 30)  
    {  
      printf("Express\n");  
    }  
    else  
    {  
      printf("Standard\n");  
    }  
  }  
  else  
  {  
    printf("Invalid Delivery Zone\n");  
  }  
  if (customerType == 2 && score >= 60)  
  {  
    printf("Premium Priority Upgrade\n");  
  }  
}  
  
return 0;  
}
```

Rubrics: Marks for each part are already given, please do follow the marking scheme, you can give 0.5 on attempt if part is not completely attempted.

You need to check the datatypes, format, specifiers, and the conditions as it is important part for the implementation of the required scenario.

Question No. 3

[20 Min, 2*5=10 Marks]

An ALPHA Hotel wants to develop a Reservation Billing System to calculate the bill for its customers. Design a pseudocode and flowchart for the following:

The hotel offers the following room types:

Choice Room Type Rate per Night

- | | | |
|---|-----------|------------|
| 1 | Standard | Rs. 5,000 |
| 2 | Deluxe | Rs. 8,000 |
| 3 | Executive | Rs. 12,000 |

For each customer, the system should take the customer's name, number of nights, and room type.

Determine the room rate according to the selected room type and calculate:

$$\text{Room Cost} = \text{Number of Nights} \times \text{Room Rate}$$

If the customer stays for 5 or more nights, give a 10% discount on the room cost. Otherwise, no discount is given.

If the customer selects an Executive Room, apply a 5% luxury tax after the discount.

Calculate the final bill:

$$\text{Final Bill} = \text{Discounted Room Cost} + \text{Luxury Tax}$$

Display the customer's name, room cost, discount, luxury tax, and final bill.

After processing a reservation, ask the user whether another reservation needs to be processed. Continue processing reservations until the user chooses to stop.

At the end, display the total revenue generated by the hotel from all reservations.

Solution:

```
CONSTANT RATE_STANDARD <- 5000.0  
CONSTANT RATE_DELUXE <- 8000.0  
CONSTANT RATE_EXECUTIVE <- 12000.0
```

```
DECLARE customerName : STRING  
DECLARE nights, roomType : INTEGER
```

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```
DECLARE roomRate, roomCost, discount, discountedCost : REAL
DECLARE luxuryTax, finalBill, totalRevenue : REAL
DECLARE continueChoice : CHARACTER
totalRevenue <- 0.0

REPEAT
  // Step 1: Input Customer Data
  OUTPUT "Enter Customer Name: "
  INPUT customerName

  OUTPUT "Enter Number of Nights: "
  INPUT nights

  OUTPUT "Select Room Type (1 - Standard, 2 - Deluxe, 3 - Executive): "
  INPUT roomType
  CASE OF roomType
    1 : roomRate <- RATE_STANDARD
    2 : roomRate <- RATE_DELUXE
    3 : roomRate <- RATE_EXECUTIVE
    OTHERWISE :
      OUTPUT "Invalid Room Type! Defaulting rate to 0."
      roomRate <- 0.0
  ENDCASE

  roomCost <- nights * roomRate

  IF nights >= 5 THEN
    discount <- roomCost * 0.10
  ELSE
    discount <- 0.0
  ENDIF

  discountedCost <- roomCost - discount

  IF roomType = 3 THEN
    luxuryTax <- discountedCost * 0.05
  ELSE
    luxuryTax <- 0.0
  ENDIF

  finalBill <- discountedCost + luxuryTax
  totalRevenue <- totalRevenue + finalBill

  OUTPUT "-----"
  OUTPUT "Customer Name: ", customerName
  OUTPUT "Base Room Cost: Rs. ", roomCost
  OUTPUT "Discount Applied: Rs. ", discount
  OUTPUT "Luxury Tax: Rs. ", luxuryTax
  OUTPUT "Final Bill Amount: Rs. ", finalBill
  OUTPUT "-----"

  OUTPUT "Do you want to process another reservation? (Y/N): "
  INPUT continueChoice

UNTIL continueChoice = 'N' OR continueChoice = 'n'
```

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```
// Step 9: Output Total Hotel Revenue
OUTPUT "===== "
OUTPUT "TOTAL HOTEL REVENUE: Rs. ", totalRevenue
OUTPUT "===== "
ENDALGORITHM
```

Rubrics:

- 1.5 marks for loop and its termination**
- 2.5 marks for proper condition checking**
- 01Marks for proper input and output generation.**

- 1.5 marks for loop and its termination**
- 2.5 marks for proper condition checking**
- 0.5 marks for proper symbols.**
- 0.5 for input output generation.**

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